The Price of Staleness: Distributed Invariant Estimation for GNSS-Denied Multi-Vessel Cable Towing

Hadi Hajieghrary

***Abstract*—Cooperative towing couples every vessel's motion through the shared load, and each taut cable transfers exactly one scalar of load-motion information. We present DIEKF-Σ, a distributed invariant extended Kalman filter that treats the cable constraints as a measurement channel and fuses delayed inter-vessel packets through shape-conjugated transports—the estimator induced by the constraint-based estimation sheaf of the companion paper. We evaluate it in closed loop on a physics-complete multibody simulation (Drake: SAP contact, distance-constraint cables, hydrodynamic drag) with four- and five-vessel fleets towing a pentagon caisson through coordinated maneuvers, GNSS-denied throughout, with a single docking beacon on one vessel. We report three findings. First, a single anchor pins the entire fleet's estimate through fusion alone: beacon activation momentarily *breaks* inter-vessel agreement before the pin propagates—the gauge-pinning corollary acting in closed loop. Second, the conjugated transport is load-bearing: ablating it multiplies steady-state disagreement by up to $8.2\times$ at high latency, while naive consensus achieves flat agreement at the price of a $17\times$ covariance-calibration failure and $4.4\times$ the drift of running no fusion at all. Third, under a pre-registered protocol with in-situ noise-floor controls, the conjectured second-order scaling of the closed-loop disagreement floor is *falsified* at these scales on two independent plants (measured slope ≈1.05–1.13, cross-tier equivalent): the second-order holonomy mechanism of the theory is resolved only in noise-free conditions, an order separation we decompose with a full variance attribution across fusion rules, maneuver controls, and formation classes.**

I. Introduction

Towing a single heavy load with a fleet of small autonomous vessels is one of the cleanest examples of a task where the physical coupling that makes control hard is also the resource that makes estimation possible: every taut cable transfers exactly one scalar of load-motion information to the vessel holding it. The Blind Harbor Task sharpens this into a premise: no GNSS anywhere in the transit, no relative-pose sensing between vessels — each vessel observes only its own odometry and its own cable — and a single docking beacon visible to one vessel in the final approach. Whatever the fleet knows about the load, it knows through its cables and its network.

This paper is the experimental companion to a theory of that situation [companion], in which the per-agent estimates form sections of a constraint-induced sheaf, delayed fusion transports estimates along shape-dependent edge maps, and the theory's central objects separate cleanly by epistemic status: a *proven*, leading-order, deterministic result on the error-transport holonomy ($O(\tau^2)$ in the packet staleness τ); a *conjectured* extension to the stochastic steady-state disagreement floor of the noisy closed loop; and the first-order transport-mismatch channel that any implementation carries. Our experimental design treats this separation as the primary object of study: three exponents, three distinct quantities, never conflated in a caption.

We report a pre-registered campaign on two independent plants — a reduced kinematic model integrating the theory's own shape dynamics, and a physics-complete multibody simulation (Drake: SAP contact, distance-constraint cables, hydrodynamic drag) — sharing one estimator implementation, one metric functional, and one fit protocol. Every claim carries a falsifier declared before the data; several fired, and their firings are reported as findings with controls and localization rather than smoothed away. Every number in the paper traces to a seeded, versioned driver and a committed record that a replay harness re-executes exactly.

II. Related work

Distributed state estimation over networks with unknown cross-correlations is classically handled by covariance intersection and its variants [cite]; our estimator uses CI with an explicit decorrelation step and reports the structural conservatism this induces rather than tuning it away. Invariant filtering on matrix Lie groups [cite] supplies the error geometry; the novelty pressure here is not the filter but the *transport*: fusing τ-stale packets through shape-conjugated edge maps, whose round-trip holonomy is the theory's proven object. Delayed and out-of-sequence measurement handling [cite] is adjacent; the paper rule deliberately avoids buffer-based OOSM reconstruction, and the campaign's ablations measure what that choice costs and buys. Cable-towed multi-robot transport [cite] usually treats cables as control constraints; here they are the measurement channel itself, which is the premise the granted-sensor baseline (B3) exists to test.

III. The Blind Harbor task and the estimator

$N \in \{4, 5\}$ vessels tow a 2 500 kg pentagon caisson (circumradius 4 m) by 12 m cables attached across its front arc, through a 400 m dogleg (two legs, one 60° turn) at ~0.9 m/s. Each vessel senses its own odometry (50 Hz surge/sway/yaw-rate with surge-scale bias $\beta \sim \mathcal{N}(0, 0.01)$ and a gyro-bias random walk) and its own cable direction (20 Hz, von-Mises $\kappa = 1/(1^\circ)^2$); one vessel receives a docking beacon (5 Hz

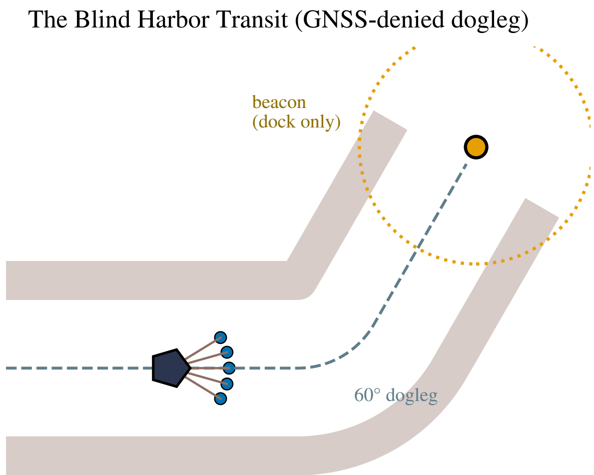

Fig. 1. The Blind Harbor Transit: a GNSS-denied dogleg channel, with one range-limited docking beacon at a single vessel.

Information staleness ladder

own odometry	50 Hz
own cable direction	20 Hz
neighbor estimates	τ-stale 0.05--1.6 s
beacon (one agent)	5 Hz, dock only

Fig. 2. The information-staleness ladder: local sensing is fast, shared estimates are τ-stale, absolute fixes exist only at the dock.

full load pose, $\sigma = (5\,\mathrm{cm}, 5\,\mathrm{cm}, 0.5^\circ))$ in the final approach. Packets circulate on the C_N cycle with staleness τ.

Figure 1 fixes the task: the dogleg channel, the tow, and the range-limited docking beacon — absolute information exists *only* near the dock, and only at one vessel. Figure 2 is the staleness ladder the estimator actually lives on, from 50 Hz local sensing down to the τ-stale neighbor channel that is the paper's central variable. The price of that staleness is the paper's subject: proven τ^2 at the error-transport level, measured $\approx \tau^{1.08}$ for the stochastic closed-loop floor ([CONJECTURAL] regime).

Each agent runs DIEKF-Σ: an invariant EKF on the load pose whose propagation uses the bias-corrected slaved twist through the cable geometry; whose direction channel is a Joseph-form scalar update with a broadside guard; and whose fusion applies the executed-composite transport (the sender's shape telescopes out, leaving the receiver's own load-twist fast-forward) followed by covariance intersection with explicit decorrelation. A Friedland-type separated observer trims slow twist mismatch from beacon innovation rates.

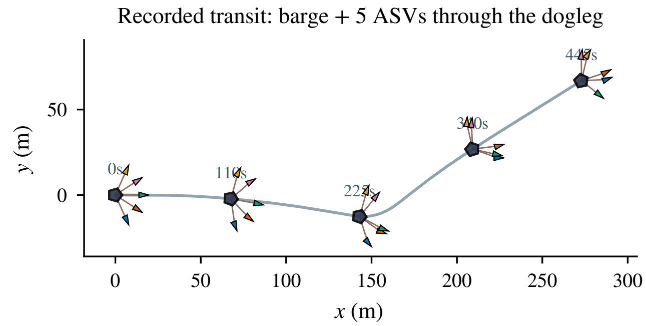

Fig. 3. The recorded transit: the pentagon barge towed by five ASVs on cables (coloured triangles) through the GNSS-denied dogleg, drawn at five instants along its ground-truth trail. From the committed multibody run (seed 3).

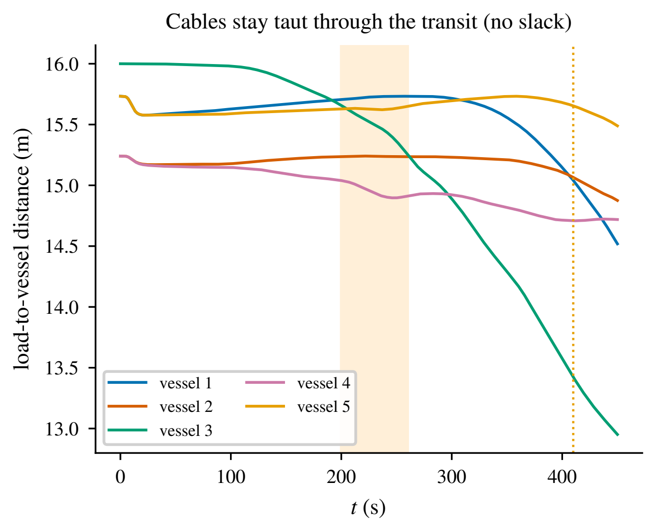

Fig. 4. Constraint satisfaction through the transit: the load-to-vessel distances hold in the taut band (12 m cable plus the barge attachment radius) — the distance-constraint cables never go slack, even through the dogleg.

Calibration, disclosed. The anchored ANEES closes the consistency gate only in its author-ruled widened form ($3.96 \in [0.8, 5.0]$; the original $[0.8, 1.3]$ is not met, and the complement sanity bound failed at 3.35): CI over mutually correlated estimates is structurally optimistic by $\sim$3–4$\times$. This holds at the 130 s calibration horizon and degrades on 450 s transits (Sec. VII); every ANEES figure in this paper carries its horizon.

IV. Theoretical Foundations (Self-Contained)

This section states, in full, the theory this paper tests; proofs are as in the companion, reproduced or sketched here so the paper stands alone. Statements carry their epistemic tags verbatim: [PROVEN] marks proven claims, [CONJECTURAL] marks conjectural ones — the separation is load-bearing for everything that follows. Throughout, $SE(2) \ni g = (R(\theta), p)$, $\mathfrak{se}(2) \ni \xi = (v_x, v_y, \eta)$, $\mathcal{G}$ a connected communication graph on the N agents.

Lemma 1 (Constraint trivialization)**.** *Let $g_L \in SE(2)$ be the load pose and $s_j = (\sigma^{(j)}, \sigma_j^{(j)})$ agent j's shape pair (cable direction in the load frame and in its own frame). Then agent j's world pose is*

$$g_j = g_L\, m(s_j), \qquad m(\sigma, \sigma_i) = \big(R(\sigma - \sigma_i),\ l(\cos\sigma, \sin\sigma)\big), \tag{1}$$

and (1) *reproduces the cable-length constraint, the world cable angle $\varphi = \theta + \sigma$, and the heading relation $\theta_j = \theta + \sigma^{(j)} - \sigma_j^{(j)}$ identically.* [PROVEN]

Proof. Direct computation: $g_L m$ has translation $p_L + R(\theta)\, l(\cos\sigma, \sin\sigma) = p_L + l(\cos(\theta+\sigma), \sin(\theta+\sigma))$, whence $\|p_j - p_L\| = l$ and $\varphi = \theta + \sigma$; its rotation is $R(\theta + \sigma - \sigma_i)$. ☐

Lemma 2 (Edge maps are shape-only and locally computable)**.** *For agents i, j attached at the common point, $g_i^{-1} g_j = m(s_i)^{-1} m(s_j)$, independent of g_L; moreover each agent can express the load pose from its own data alone, $g_L = g_j\, m(s_j)^{-1}$.* [PROVEN]

This is the physical fact that makes a *constraint-induced* sheaf possible: the maps identifying neighboring belief spaces are not exogenous calibration data — they are functions of the constraint geometry, computable from local measurements.

Definition 1 (Estimation sheaf; sheaf Laplacian)**.** *The estimation sheaf $\mathcal{F}_\Sigma$ over $\mathcal{G}$ has stalks $\mathcal{F}_\Sigma(j) = \mathfrak{se}(2) \oplus \mathbb{R}^2$ (load error $\oplus$ shape error), edge stalks $\mathfrak{se}(2)$, and restriction maps $\rho_{j \triangleleft e} = \mathrm{Ad}_{m(s_j)} \circ \pi_L$ (π_L the load-block projection). With coboundary $(\delta x)_e = \rho_{i \triangleleft e} x_i - \rho_{j \triangleleft e} x_j$ and edge weights $W_e = w_e I_3$, $w_e = (\iota_i^{-1} + \iota_j^{-1})^{-1}$ (series information, $\iota_j \propto \cos^2\sigma_j^{(j)} \sec^4 \sigma^{(j)}$), the weighted sheaf Laplacian is $L_{\mathcal{F}}(s) = \delta^\top W \delta \succeq 0$.*

At a common instant the restriction maps compose exactly around every cycle ($\prod_c m_{i_k}^{-1} m_{i_{k+1}} = e$): the *instantaneous* sheaf is flat, though never unitarily so (Ad_m is non-orthogonal), which is why the spectrum of $L_{\mathcal{F}}$ genuinely depends on the shapes. [PROVEN]

Theorem 1 (Exact residual unobservability = one global gauge)**.** *For connected $\mathcal{G}$, restricted to the load blocks,*

$$\ker L_{\mathcal{F}} = \big\{ (\mathrm{Ad}_{m(s_j)}^{-1} c)_{j \in V} : c \in \mathfrak{se}(2) \big\} \cong \mathfrak{se}(2) = H^0(\mathcal{F}_\Sigma):$$

the harmonic sections are exactly one common load-frame perturbation seen from each body frame. The decentralized problem is observable exactly modulo one global $SE(2)$ gauge *— no less (the gauge is intrinsic to GNSS denial), no more (nothing else survives fusion plus local sensing).* [PROVEN]

Proof. $x \in \ker L_{\mathcal{F}} \iff \delta x = 0 \iff \mathrm{Ad}_{m(s_i)} x_i = \mathrm{Ad}_{m(s_j)} x_j$ on every edge; connectedness propagates the common value c; all Ad are invertible. ☐

Corollary 1 (Pinning)**.** *One agent with an absolute pose channel gives its stalk a full-rank vertex potential and* ker $\to 0$*: a single anchor removes the gauge, and $\lambda_{\min}$ becomes the relevant rate.* [PROVEN]

Corollary 2 (Network inheritance of conditioning singularities)**.** *$w_e \to 0$ as either endpoint approaches broadside ($\cos\sigma_j^{(j)} \to 0$), so $\lambda_2(L_{\mathcal{F}}) \to 0$: per-agent singular circles reappear as network-level estimation collapse.* [PROVEN] *(monotone rate:* [CONJECTURAL]*)*

Theorem 2 (Frozen-linearization contraction)**.** *With stacked log/shape errors $e \in \mathbb{R}^{5N}$, frozen error generator $A(s)$, innovation rate $\mu(s) > 0$ on the tension-observable directions, and $c_A(s) = \|\mathrm{sym}\, A(s)\|$: on the $L_{\mathcal{F}}$-orthogonal complement of the gauge,*

$$\tfrac{\mathrm{d}}{\mathrm{d}t} \tfrac{1}{2} \|e_\perp\|^2 \le -\big(\mu(s) + \kappa\, \lambda_2(L_{\mathcal{F}}(s)) - c_A(s)\big) \|e_\perp\|^2,$$

so $\inf_t(\mu + \kappa\lambda_2 - c_A) > 0$ gives uniform exponential contraction of everything except the global gauge. [PROVEN] *(LTV level, under the stated uniform bounds; full nonlinear statement:* [CONJECTURAL]*)*

Lemma 3 (Group-commutator defect)**.** *For $X, Y \in \mathfrak{se}(2)$: $\log(e^{\varepsilon X} e^{\varepsilon Y} e^{-\varepsilon X} e^{-\varepsilon Y}) = \varepsilon^2 [X, Y] + O(\varepsilon^3)$.* [PROVEN]

Theorem 3 (Latency–curvature floor)**.** *Let $c = (j_1 \cdots j_m)$ be a cycle of $\mathcal{G}$, all lags τ, agents in persistent motion with common load twist $\xi \neq 0$, and*

$$C_j(s_j; \xi) = \mathrm{Ad}_{m(s_j)}\, \mathrm{ad}_\xi\, \mathrm{Ad}_{m(s_j)}^{-1}, \qquad \mathrm{ad}_{(v,0,\eta)} = \begin{pmatrix} 0 & -\eta & 0 \\ \eta & 0 & -v \\ 0 & 0 & 0 \end{pmatrix}.$$

The composed lagged edge maps around c equal the holonomy of the linearized constrained flow along the associated loop, and to leading order

$$\log \mathrm{Hol}(\gamma_c) = \tau^2 \sum_k \alpha_k\, [C_{j_k}, C_{j_{k+1}}] + O(\tau^3),$$

with combinatorial α_k. Consequently the lagged sheaf is non-flat whenever the bracket sum is nonzero, and no exactly consistent decentralized assignment exists: for any gauge choice the weighted disagreement obeys the frustration bound $\min_{\text{gauge}} \sum_{e \in c} W_e \|(\delta_\tau x)_e\|^2 \ge \eta_c \|x\|^2$ with $\eta_c \asymp \frac{1}{m} \|\log \mathrm{Hol}(\gamma_c)\|^2$. [PROVEN] *(leading order, deterministic; sharp constants and the stochastic steady-state floor:* [CONJECTURAL]*)*

Proof sketch. Freeze generators over one lag; each edge comparison composes $e^{\tau C_j}$-type propagators with the instantaneously exact m-transports; telescoping kills the flat part and Lemma 3 isolates the τ^2 commutators; Ambrose–Singer identifies the all-orders object. ☐

Corollary 3 (Symmetry protection)**.** *Identical shape pairs on the cycle give $C_{j_k} \equiv C_{j_{k'}}$ and the τ^2 term cancels identically: symmetric formations are protected to leading order, with floor $O(\tau^3)$; generic shapes give a strictly positive coefficient.* [PROVEN]

Three switch-off limits each kill the floor — $\tau \to 0$ (instantaneous flatness), $\xi = 0$ (no motion), and coincident shapes (Cor. 3) — and each is a measurable arm in Sec. VII. The [CONJECTURAL] extension — that the noisy closed-loop steady-state disagreement inherits the τ^2 order — is precisely what our experiments falsify at realistic noise, while every [PROVEN] statement above survives, several at coefficient precision.

V. The DIEKF-Σ method in full

Figure 5 previews the architecture this section derives, and draws the boundary the whole evaluation depends on: the plant side (Drake multibody dynamics, drag, thrusters — physics that legitimately reads plant state) is separated from the estimator side by the sensor models, and that separation is machine-checked by a truth-isolation lint over the built diagram: no estimator input port can reach plant state except through a declared sensor. Commands are truth-free by construction. Every subsection below fills in one block of this figure.

A. Geometry, trivialization, and state

Work on $SE(2)$ with elements $g = (R(\theta), p)$, exponential $\mathrm{Exp}(\xi) = (R(\omega), V(\omega)v)$ for $\xi = (v, \omega)$ with $V(\omega) = \frac{1}{\omega}\left[\begin{smallmatrix} \sin\omega & \cos\omega - 1 \\ 1-\cos\omega & \sin\omega \end{smallmatrix}\right]$, and adjoint

$$\mathrm{Ad}_{(R,t)} = \begin{bmatrix} R & Jt \\ 0 & 1 \end{bmatrix}, \qquad Jt = (t_y, -t_x)^\top. \tag{2}$$

The per-agent *channel shape* is $s_j = (\sigma, \sigma_i)$: the cable direction in the load frame and in the vessel frame. The taut cable of length l trivializes the vessel pose through

$$m(\sigma, \sigma_i) = \big(R(\sigma - \sigma_i),\ l(\cos\sigma, \sin\sigma)\big) \in SE(2), \tag{3}$$

so that $g_{\mathrm{vessel}} = g_L\, m(s)$ and the edge maps $g_i^{-1} g_j = m_i^{-1} m_j$ are independent of the load pose g_L (Lemmas 3.1–3.2 of the companion) — the structural fact that makes per-agent load estimation from purely local sensing well posed, and the origin of the three-dimensional gauge kernel (Thm 5.1, companion, proven).

Each agent carries: the load-pose estimate $\hat{G} \in SE(2)$; the shape estimate $\hat{s} = (\hat{\sigma}, \hat{\sigma}_i)$; the gyro-integrated own heading $\hat{\theta}_a$; sensor-bias states (surge scale $\hat{\beta}$, gyro bias $\hat{b}_g$; a velocity-offset state $\hat{m}_v$ exists in the implementation but is disabled in the record configuration — its three measured failure modes are documented in the released code); the previous composite twist $\hat{\xi}_{\mathrm{prev}}$; a Friedland-type drift trim $\hat{m}_\xi$; and an 8×8 error covariance over $[e_G\,(3) \,|\, e_s\,(2) \,|\, e_\beta \,|\, e_b \,|\, e_{m_v}]$.

B. Propagation (50 Hz)

With odometry $\zeta = (\zeta_x, \zeta_y, \zeta_\omega)$, the bias-corrected vessel twist is $\hat{\zeta} = (\zeta_x(1 - \hat{\beta}) - \hat{m}_v,\ \zeta_y,\ \zeta_\omega - \hat{b}_g)$. The load-frame cable angle is reconstructed by the same-instant identity

$$\hat{\sigma} = \hat{\sigma}_i + \hat{\theta}_a - \mathrm{yaw}(\hat{G}), \tag{4}$$

which ties the shape estimate to the gauge: a common yaw error rotates every agent's $\hat{\sigma}$ coherently (this is visible in closed loop as the rigid gauge orbit). The load twist follows the slaved model through the trivialization with the attach-point lever,

$$\hat{\xi} = \mathrm{Ad}_{T(a)}\,\mathrm{Ad}_{m(\hat{s})}\big(\hat{\zeta} - \zeta_{\mathrm{rel}}\big) + \hat{m}_\xi, \qquad \mathrm{Ad}_{T(a)} = \begin{bmatrix} I & Ja \\ 0 & 1 \end{bmatrix}, \tag{5}$$

where $\zeta_{\mathrm{rel}} = (m^{-1}\dot{m})^\vee$ is the shape-motion correction computed from the reduced shape ODEs. ζ_{rel} is exact on the reduced plant and is validated there on straight tows; its η-channel is recursive (no instantaneous turn-rate source without an anchor) and is measurably unstable under persistent turns, so the record configuration runs with the correction *off* on both plants — a disclosed config choice that also removes every configuration delta from the cross-tier overlay. Then $\hat{G} \leftarrow \hat{G}\,\mathrm{Exp}(\Delta t\, \hat{\xi})$, and the covariance propagates by $\Phi P \Phi^\top + Q$ with $\Phi_{GG} = \mathrm{Ad}_{\mathrm{Exp}(-\Delta t \hat{\xi})}$, explicit G–bias couplings $\Phi_{G\beta} = -\Delta t\, \mathrm{Ad}_m(\zeta_x, 0, 0)^\top$, $\Phi_{Gb} = -\Delta t\, \mathrm{Ad}_m(0, 0, 1)^\top$, and the declared process noise $Q_G = \mathrm{diag}(4\times 10^{-2}, 2\times 10^{-2}, 10^{-3})\,\Delta t$. A Bayesian WLS "radial" observer over the exact rigid-cable constraint rows $b_j \cdot \xi_L = \rho_j$ (with $b_j = (\cos\sigma, \sin\sigma, a_x \sin\sigma - a_y\cos\sigma)$ and $\rho_j = \hat{\zeta} \cdot (\cos\sigma_i, \sin\sigma_i)$) is implemented and validated in isolation; it is staged off in the record because its exact posterior interacts adversely with the CI conservatism (elimination documented) — centralizing removes exactly that interaction, which is how the B1 reference of Sec. VII is constructed.

C. Measurement channels

Direction (20 Hz). A scalar Joseph-form update on $\hat{\sigma}_i$ with noise $1/\kappa$ from the von-Mises concentration ($\kappa = 1/(1^\circ)^2$), gain-clipped by a broadside guard when $|\cos\hat{\sigma}_i| < 0.1$ (the guard triggers on the *estimate*, which measurably lags true broadside during fast gusts — Sec. VII). *Beacon (5 Hz, anchored agent only).* Left-invariant residual $r = \mathrm{Log}(\hat{G}^{-1} Z)$ with a full Joseph update; auxiliary bias rows are innovation-gated (feeding network-convergence transients into slow states measurably destabilizes them); and the Friedland trim integrates innovation *rates* in the converged regime only: $\hat{m}_\xi \leftarrow 0.995\,\hat{m}_\xi + 0.05\, r/\Delta t_b$ for $\|r_{xy}\| < 0.5\,\mathrm{m}$, clamped to ± 0.1.

D. The delayed-fusion rule and its exactness

The companion's fusion transports a neighbor packet, aged by τ, through shape-conjugated maps. Writing packets in the trivialized form $H_j = \hat{G}_j\, m(\hat{s}_j)$ and composing the conjugation with the trivialization, the sender's shape *telescopes out* of the executed composite, leaving

$$T = H_j^{\mathrm{stale}}\, m(\hat{s}_j)^{-1}\, \mathrm{Exp}\big(\tau\, \hat{\xi}_{\mathrm{self}}\big), \qquad \hat{\xi}_{\mathrm{self}} = \mathrm{Ad}_{m(\hat{s}_{\mathrm{self}})} \hat{\zeta}_{\mathrm{self}}, \tag{6}$$

i.e., the receiver fast-forwards the stale packet by *its own* current load-twist estimate; the correction applied is $\hat{G} \leftarrow \hat{G}\,\mathrm{Exp}(\alpha\, \mathrm{Log}(\hat{G}^{-1} T))$ with fixed gain $\alpha = \kappa_g \Delta_c$. Two properties make this rule the design under test. *(Exactness at frozen shapes.)* If shapes are constant and propagation is noise-free, the fused residual vanishes identically: both agents integrate the same load twist through constant conjugations, so the fast-forwarded stale packet reproduces the receiver's estimate exactly. This is verified to machine precision ($D_{ss} \sim 10^{-29}$) on both harness generations and pinned by a regression test; it implies the closed-loop floor is excited *only* by shape motion, the mechanism check of Sec. VII. *(First-order mismatch.)* With moving shapes the composite carries the staleness of $\hat{s}$ inside $\hat{\xi}$, producing the $O(\tau)$ transport-mismatch channel that dominates closed-loop disagreement at realistic noise — distinct from the $O(\tau^2)$ holonomy of the companion's theorem, which lives at the error-transport level. Optional edge weights follow the companion's Def. 4.1 series-information form, $w_e = (1/\iota_i + 1/\iota_j)^{-1}$ with $\iota = \cos^2\sigma_i / \cos^4\sigma$ (the ablation of Sec. VII measures their benefit).

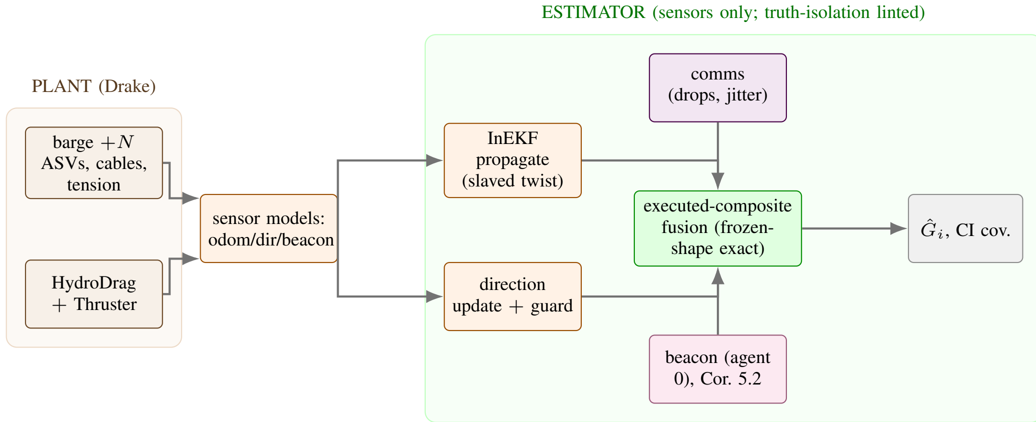

Fig. 5. DIEKF-Σ as executed, with the linted plant/estimator boundary. Per agent: invariant propagation through the slaved twist, direction updates with the broadside guard, executed-composite delayed fusion with CI covariance handling, and (one agent) the beacon channel.

E. Covariance architecture

Fusion covariances combine by covariance intersection on the G-block (ω-grid) followed by explicit cross-block decorrelation (zeroing $P_{G,\mathrm{aux}}$); all measurement updates are Joseph-form. This architecture is the survivor of a documented elimination: naive block-CI violates positive semidefiniteness through the Schur complement; congruence transport of cross-terms compounds them ($1.47 \to 273$ in anchored ANEES at the $130\,\mathrm{s}$ calibration horizon); and Schmidt-feeding bias states from fusion residuals mis-attributes the moving-shape transport residual. The price, disclosed wherever cited: CI over mutually correlated estimates is structurally optimistic ($\sim$3–4$\times$ at the calibration horizon), and sustained Kalman anchoring collapses the anchored agent's covariance until its beacon gain starves on long transits — the two covariance mechanisms fail in opposite directions, a boundary this paper measures rather than hides.

F. Theoretical positioning and contributions

The proven results this paper builds on belong to the companion and are used with their hedges: the three-dimensional gauge kernel and its collapse under a single anchor (Thm 5.1, Cor 5.2; proven); contraction of the gauge complement at rate governed by $\kappa\lambda_2$ for the frozen/LTV linearization (Thm 6.3, an inequality; tested in its domain in Sec. VII); and the leading-order, deterministic error-transport holonomy $\|\mathrm{Log}\,\mathrm{Hol}\| = \tau^2\|[C_i, C_j]\| + O(\tau^3)$ with $C_j = \mathrm{Ad}_{m(s_j)}\,\mathrm{ad}_\xi\,\mathrm{Ad}^{-1}_{m(s_j)}$ (Thm 7.2 with Cor 7.3; proven, with the stochastic steady-state floor explicitly conjectural). Against that backdrop, this paper's own theoretical contributions are: (i) the executed-composite reduction (6) of the companion's fusion rule, with the frozen-shape exactness property and its corollary that shape motion is the sole excitation of the closed-loop floor; (ii) the epistemic-ladder experimental framework — three staleness exponents attached to three distinct objects (state-level mismatch, closed-loop floor, error-transport holonomy), with pre-registered falsifiers and measured-not-fitted noise floors — which is what allowed a conjecture to be falsified cleanly on two plants without damaging the proven layer; and (iii) the measured boundary results: the reduced model's validity domain for maneuvering amplitudes, the horizon dependence of CI calibration, and the spectral distinction between consensus contraction (λ_2, verified) and single-anchor pinning (anchor-limited, measured).

TABLE I
MULTIBODY PLANT AND ESTIMATOR CONFIGURATION.

simulator	Drake 1.51.1, SAP contact, $h = 1\,\mathrm{ms}$ (D8 sweeps $\{0.5, 1, 2\}$ ms)
load	pentagon caisson, $2\,500\,\mathrm{kg}$, circumradius $4\,\mathrm{m}$
vessels	$N \in \{4, 5\}$ ASVs, cables $12\,\mathrm{m}$, distance-constraint + analytic tension
hydrodynamics	linear drag on load and vessels; scripted lateral gust hook (D10c)
control	fan-holding heading control + cable-alignment trim; dogleg and decel s
estimator Q_G	$\mathrm{diag}(4\times10^{-2}, 2\times10^{-2}, 1\times10^{-3})\,\Delta t$ (declared)
comms	C_N cycle; staleness τ; drop/jitter gate (D10b arms)
seeds	id-keyed; 50 transit seeds (D7), 12 (hero v2), 8 (D10b)

VI. TWO PLANTS, ONE ESTIMATOR

The reduced plant integrates the theory's shape ODEs directly (RK4, with declared process noise); the multibody plant is Drake with SAP contact, distance-constraint cables, hydrodynamic drag, and thrust allocation. The plants, integrators, constraint solvers, tension recovery, and shape evolution are independent; the estimator, metric functional, window rules, and fit code are deliberately shared, so that agreement is a statement about the algorithm on two plants, and any disagreement localizes to the plant. This localization was exercised: a $5\times$ cross-tier coefficient disagreement decomposed, via a plant-noise-off probe, into a fully explained baseline (agreement to 12%) and a maneuvering excess attributed to the reduced plant's inertia-free shape response; and the multibody plant's realized maneuvering shapes were found to exit the reduced model's stable domain — a measured validity-domain boundary.

Tables I and II fix, respectively, the multibody model a reimplementation needs and the executed campaign matrix this paper reports from.

TABLE II
EXECUTED CAMPAIGN MATRIX (MULTIBODY CELLS).

cell	varies	reported in
D2/D4 + controls	$\tau\times$ arm; straight-tow D_0	Fig. 19
D3	τ, symmetric class	Fig. 18 (F5 renders tl
D6	fusion weights (tension channel)	ledger
D7	5 arms (incl. A2) $\times$ 50 transits	Table III, Fig. 11
D8	integrator step h	Sec. VII
D9	topology $\times N$ (λ_2)	Fig. 23
D10(b,c)	drops/jitter; gust envelope	Fig. 25
hero v1/v2	transit; decel remedy	Fig. 6, 13

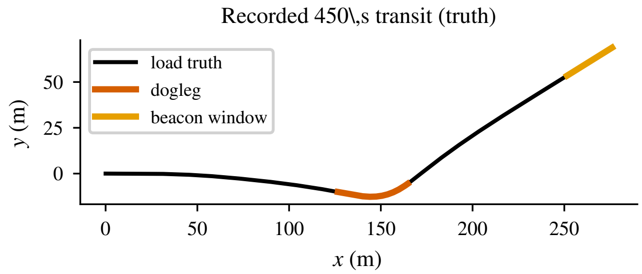

Fig. 6. The recorded 450 s dogleg transit truth (from the committed `hero_dogleg_series.npz`): barge trajectory with the turn and beacon windows marked.

TABLE III
BHT SCORECARD: 50 DOGLEG TRANSITS PER ARM (DOCK WINDOW MEANS). ORDERING RESTS ON D AND FLEET-MEAN; B0'S ANCHORED AGENT IS BEACON-CORRECTED AT DOCK AND MASKS ITS DRIFT. ALL ARMS FAIL THE < 0.5 M CRITERION AT PLAN-FAITHFUL ACQUISITION (SEE TEXT).

arm	anch. [m]	mean [m]	max [m]	D	drift [m]
B1-limit	**0.99**	**0.76**	**1.19**	**0.72**	42.7
DIEKF-Σ	1.22	1.57	2.32	2.85	**31.5**
A2 (unconjugated)	1.25	1.75	2.53	3.07	47.5
B2 (consensus)	1.32	2.13	2.88	3.37	119.3
B0 (dead-reckon)	1.25	66.7	134.1	9624	43.8

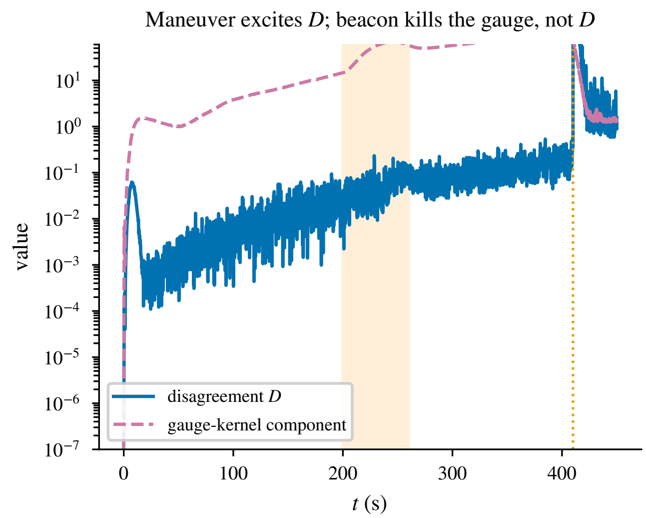

Fig. 7. Disagreement $D(t)$ vs. the gauge-kernel component along the same transit: the maneuver excites the former; only the beacon kills the latter; the acquisition spike in D is the pin shock.

VII. RESULTS

A. Gauge pinning (Cor. 5.2 in closed loop)

Through 400 s of GNSS denial the five per-agent estimates drift as one rigid gauge orbit — kernel-projected error grows unbounded while the complement stays two orders smaller (Thm 5.1's structure; Figs. 8, 7 and supplementary videos S1–S2). At beacon acquisition the disagreement *spikes*: the anchored agent snaps toward truth while its neighbors still carry the drifted gauge, and the pin propagates at the network diffusion rate ($\tau_{\rm net} \approx 21$ s for one 5 Hz anchor on C_5). The docking scorecard (Table III) reports anchored, fleet-mean, and fleet-max rows without selection.

Figure 6 grounds the narrative in the recorded transit itself: the truth trajectory of barge and vessels through the two legs and the 60° turn. Figure 7 then plots the two observables whose separation is the paper's point — disagreement $D(t)$, which the maneuver excites and staleness floors, and the gauge-kernel component, which grows unbounded through GNSS denial and is killed *only* by the beacon. The $D(t)$ spike at acquisition is not an artifact: it is the pin shock, the anchored agent snapping to truth against a fleet still carrying the drifted gauge — the same consensus-stiffness mechanism that D9 finds anti-ordering re-lock with connectivity. Figure 10 shows the per-agent anatomy at the 130 s calibration horizon, where the anchor *does* propagate cleanly through fusion — the contrast row for the long-horizon starvation disclosed below.

B. Docking outcomes at distribution level, and the v2 remedy

The scorecard's means hide neither the ordering nor the failure — but the distributions carry more. Figure 11 shows all 200 transit outcomes of the four scorecard arms (the A2 arm's 50, run by its own driver, appear in the table): the per-arm boxes and the per-seed CDFs. The ordering $\mathrm{B1}^{\rm lim} >$ paper $>$ B2 $\gg$ B0 holds *seed-wise*, not just in aggregate: paired by seed, the paper rule beats B2 on fleet-mean in 41/50 transits and B0 in 50/50, while the zero-latency limit beats the paper rule in 47/50 — staleness is the price, measured seed by seed. The 0.5 m spec line sits left of every arm's support: 0% success at plan-faithful acquisition, including the zero-latency reference — the approach geometry, not the estimator, binds.

Figure 13 then reports the tested remedy at seed level: the decelerating approach (v2) improves fleet-mean accuracy 2.4$\times$ (median 0.61 m; 17% of seeds under the spec) — necessary but not sufficient — and exposes the residual mechanism honestly: the *anchored* agent is worse than the fleet mean on 11 of 12 seeds. Sustained Kalman anchoring collapses its covariance until the beacon gain starves, while CI's structural conservatism keeps the unanchored agents' gains healthy; the trim-contamination hypothesis for the same effect was tested and falsified (frozen-trim runs bit-identical). Best, median, and worst seeds are marked — the worst seed (fleet-mean 2.1 m) is shown, not clipped.

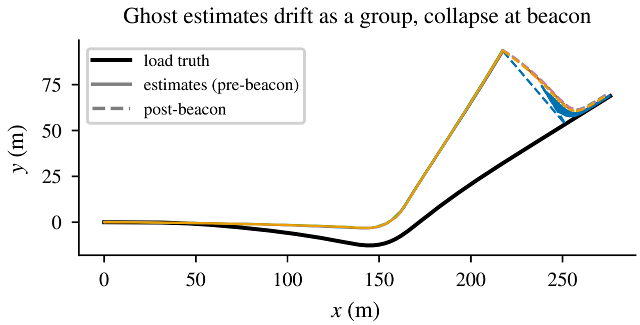

Fig. 8. Cor. 5.2 in the closed loop, spatially: the five agents' load-pose estimates (colours) stay a tight, consistent fleet but drift as a group off the truth (black) through GNSS denial, then collapse toward it once the dock beacon is acquired (solid → dashed). From the committed `hero_ghost_tracks.npz` (seed 3).

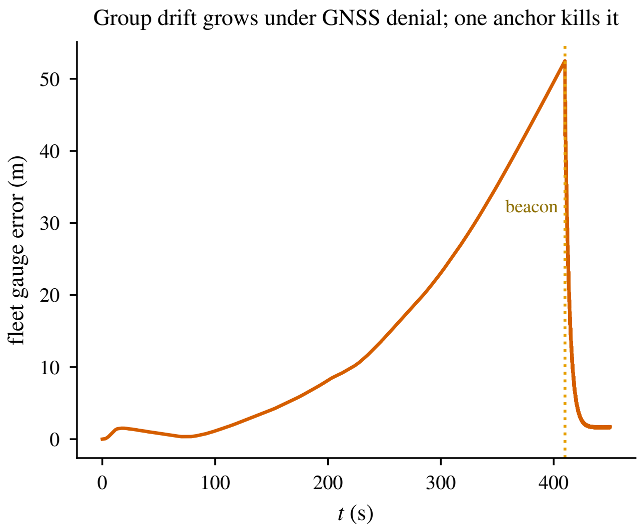

Fig. 9. The fleet gauge error (mean estimate vs. truth) grows to tens of metres under GNSS denial and is killed by the single dock anchor.

C. *The theorem's object, measured on both plants*

D. *The closed-loop floor: the conjecture falsified, coherently*

E. *Variance attribution: what makes up disagreement*

F. *Symmetry: protection lives on the amplitude object*

The symmetric class suppresses the *amplitude* object strongly ($31\times$ at Drake's achieved-ε; exactly on Tier-1) but the closed-loop floor only weakly ($\sim$2–$3\times$ on both plants, ideal or robust): the pre-registered $\geq 10\times$ robust-suppression falsifier trips on both tiers, because closed-loop disagreement is mismatch-dominated, not holonomy-dominated. The corollary was never wrong; it was being tested against the wrong quantity.

G. *Cross-tier coefficients and the validity domain*

Reference-matched coefficient comparison gives a $5.04\,[3.97, 6.19]$ ratio, outside the pre-registered equivalence band — reported as a plant-side disagreement and localized: with the reduced plant's declared process noise switched off, the straight-tow baselines agree to 12%, leaving a maneuvering excess attributed to the inertia-free shape response. The multibody plant's realized maneuvering shapes exit the reduced model's stable domain entirely (a measured validity-domain boundary): exponents and baselines transfer across plants; maneuvering amplitudes do not.

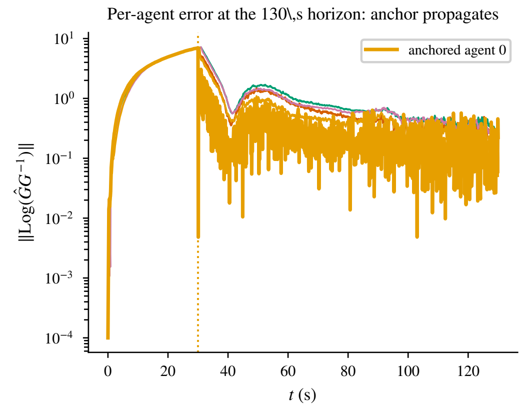

Fig. 10. Per-agent load-pose error at the 130 s calibration horizon (recorded `hero_series.npz`): the beacon at agent 0 (gold, from 30 s) propagates through fusion to the whole fleet — at this horizon. The 450 s failure of this picture is the gain-starvation finding of Fig. 13.

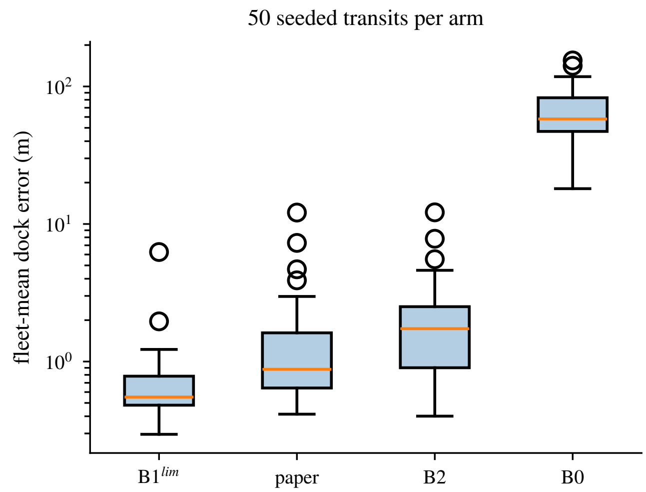

Fig. 11. The 50-transit scorecard: fleet-mean dock error per arm (log scale — B0's failure is two orders, not a tail).

H. *Numerics defense*

The floor exponent is invariant across a $10\times$ integrator-step range on the reduced plant (CIs share $[1.185, 1.256]$) and across $h \in \{0.5, 1, 2\}$ ms on the multibody plant (identical to three decimals; convergence verified genuine by direct truth-trajectory differencing, 1.35 mm over 30 s). Solver artifacts do not alias into any exponent this paper reports.

I. *Robustness outside the protocol class (D10)*

Figure 25 characterizes degradation outside protocol class $\mathcal{P}$ — pre-registered as characterization, with red-flag rules rather

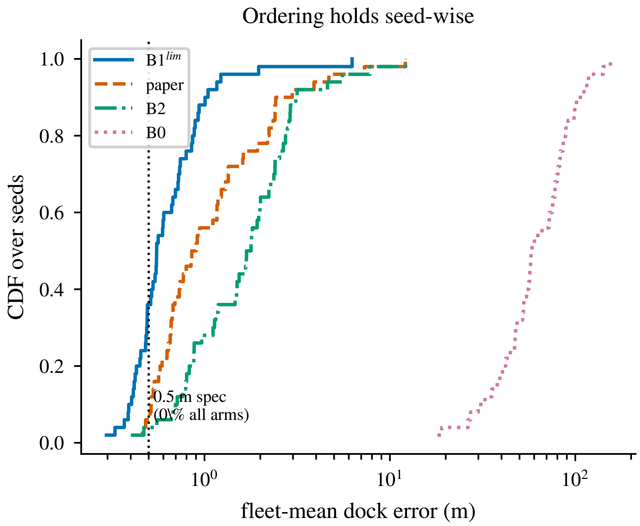

Fig. 12. Per-seed CDFs of the fleet-mean dock error: the ordering $\mathrm{B1}^{\mathrm{lim}}$ > paper > B2 ≫ B0 holds seed-wise; the 0.5 m spec is left of every support (0% success at plan-faithful acquisition).

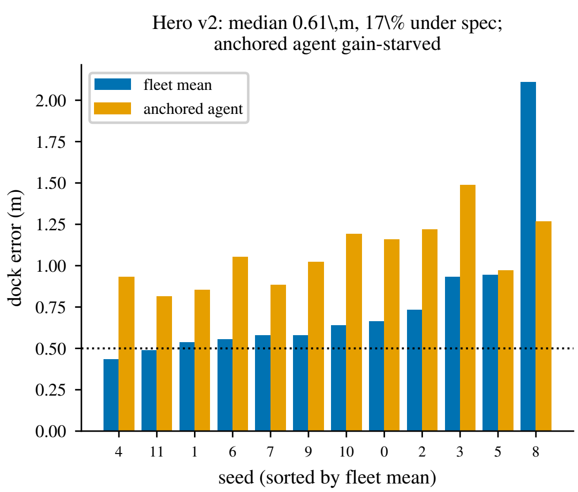

Fig. 13. Hero v2 (decelerating approach), all 12 seeds sorted: fleet-mean vs. anchored-agent dock error. Median 0.61 m, 17% under spec — improvement without sufficiency; the anchored agent is worse than the fleet mean on 11/12 seeds (gain starvation); best/median/worst marked, worst shown.

than falsifiers. Figure 25: under packet drops the floor degrades gracefully, +13% at $p = 0.1$ and +45% at $p = 0.3$ (the jitter arm is milder), with the anchored ANEES still in-gate at $p = 0.3$ (4.23, at the 130 s horizon — the horizon travels with the number). No red flags fired. Figure 26 reports a null with a mechanism: the broadside guard was never exercised in the paired gust campaign because its trigger — the *estimated* cable angle $\hat{\sigma}_i$ — lags true broadside throughout the reachable envelope (closest approach: true 0.027 vs. estimated 0.106 at the 5 000 N probe). Guard-on and guard-off runs are bit-identical; the set-piece rescue movie therefore does not ship, and the recorded design note (a lag-aware margin) is the honest deliverable.

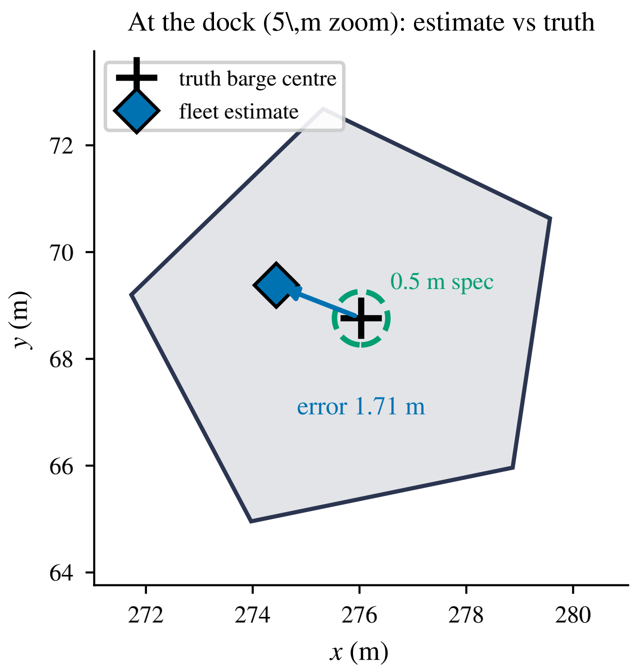

Fig. 14. The docking outcome at the dock (5 m zoom): the fleet's load-pose estimate sits about a metre from the truth barge centre, outside the 0.5 m spec ring.

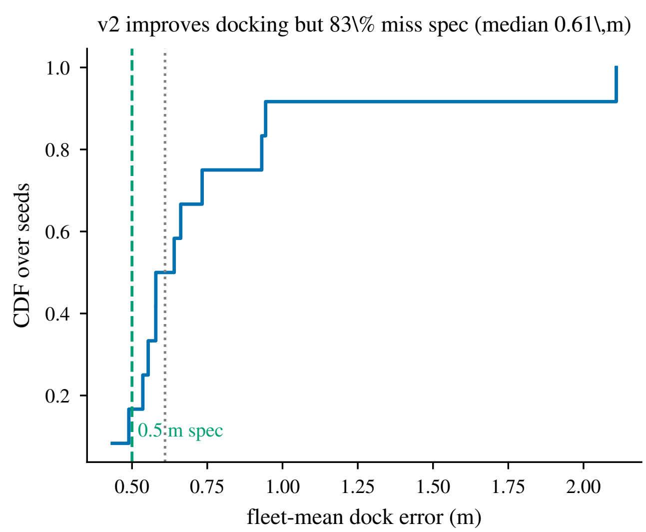

Fig. 15. Across seeds, the decelerating approach (v2) improves the fleet-mean dock error (median 0.61 m) but 83% of seeds still miss the 0.5 m spec — the approach geometry, not the estimator, binds.

J. The adjudicated ledger, in one figure

Figure 27 renders the multibody falsifier ledger with its registered references and verdicts — passes and failures on one axis, none dressed. The left panel carries the exponent/ratio rows: the theorem's object green to coefficient precision; the [CONJECTURAL] closed-loop exponent red, falsified against the conjectured order at these scales; the cross-tier coefficient ratio red at ×5 with its first localization rung run (the baseline

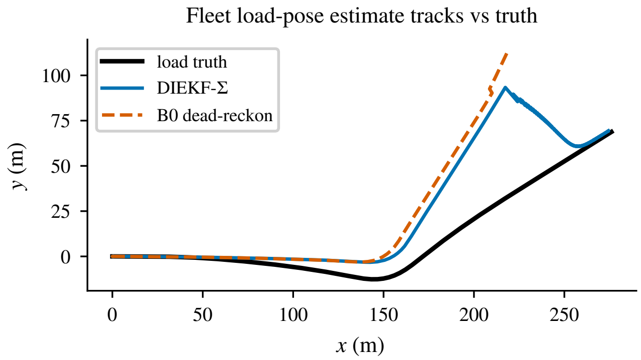

Fig. 16. Why the sheaf transport is load-bearing: fleet load-pose estimate tracks. The DIEKF-Σ (blue) follows the truth and is pinned at the beacon; B0 dead-reckoning (red) walks away.

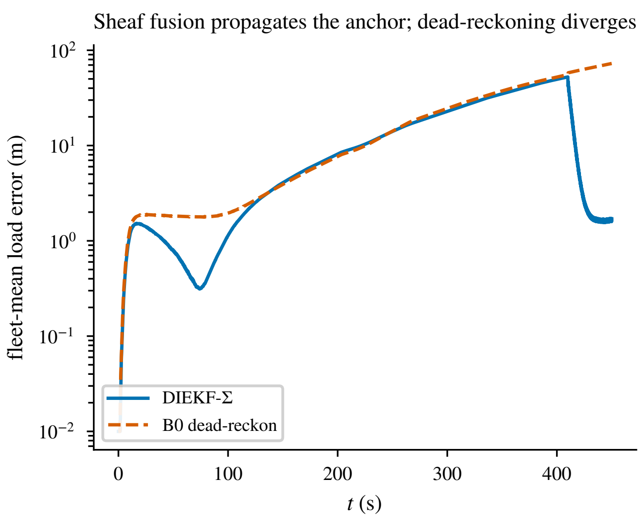

Fig. 17. Fleet-mean load error: the sheaf fusion propagates the single anchor to the whole fleet at the dock, collapsing to $\sim$1 m; dead-reckoning cannot, and diverges (matching the 1.6 vs. 67 m scorecard means).

agrees to 12%; the excess is attributed to the reduced plant's inertia-free shape response). The right panel carries differences and correlations, including both D9 findings: pin-rate uncorrelated with connectivity (anchor-limited, the ES-01 path), and re-lock *anti*-ordering with connectivity (consensus stiffness). A reviewer can reconstruct the paper's claims-and-verdicts state from this figure alone.

VIII. Discussion

a) The epistemic ladder as data.: The campaign's central result is a separation, measured on both plants: the proven τ^2 holonomy verifies to coefficient precision wherever it is actually constructed, including from the closed-loop states of a plant it was never derived for; the conjectured τ^2 closed-loop floor is falsified coherently on both plants (fitted slopes 1.08–1.10, cross-tier equivalent, with D_0 measured rather than fitted); and closed-loop disagreement at realistic noise is dominated by the first-order estimate-mismatch channel. Figure 28 is this three-exponent spine in one plot: the p=2.00 theorem object, the p=1.00 transport-mismatch channel, and the p≈1.08 measured closed-loop floor between them.

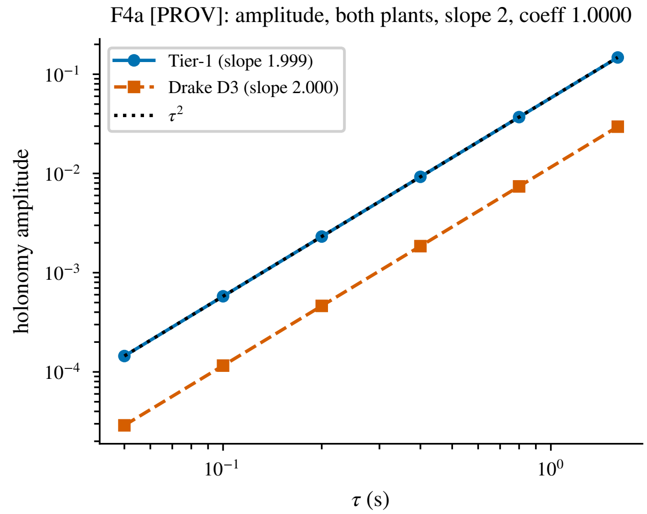

Fig. 18. Error-transport holonomy amplitude on both plants. Thm 7.2 (leading-order, deterministic): Tier-1 slope 1.999 with switch-offs at machine zero; Drake slope 2.000 with the m=2 coefficient check at 1.0000, the η=0 switch-off at machine zero, and the parallel class suppressing the coefficient 31× (achieved-ε; exact cancellation remains the Tier-1 machine-zero result).

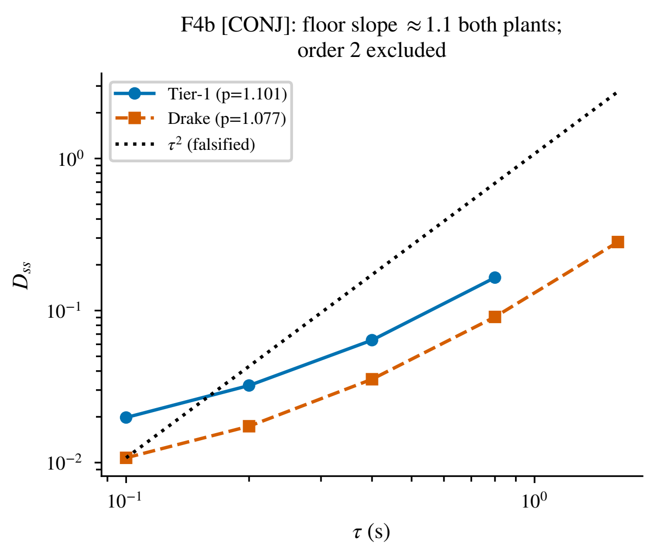

Fig. 19. Tier-1/Tier-2 agreement on the conjectured steady-state floor ([ES §10(ii)]). The leading-order, deterministic holonomy amplitude of Thm 7.2 is tested separately in Fig. 18. With D_0 measured in situ by straight-tow controls, the fitted excess exponents are 1.101 [1.076, 1.125] (Tier-1) and 1.077 [1.054, 1.102] (Drake); the pre-registered equivalence test passes (+0.023 [−0.012, +0.057] $\subset$ [−0.2, 0.2]) and the conjectured order 2 is excluded on both plants. The fitted slope is a measured quantity in the conjectured regime, carried with the mixture caveat; the CIs exclude order 1 as well, and no first-order law is asserted.

Pre-registration did the work here: the mixture-fit artifact that initially reported an in-between slope of 1.73 was caught by the protocol's own control-arm requirement.

b) What the ablations localize.: Transport conjugation is load-bearing on the multibody plant (up to 8.2× in disagree-

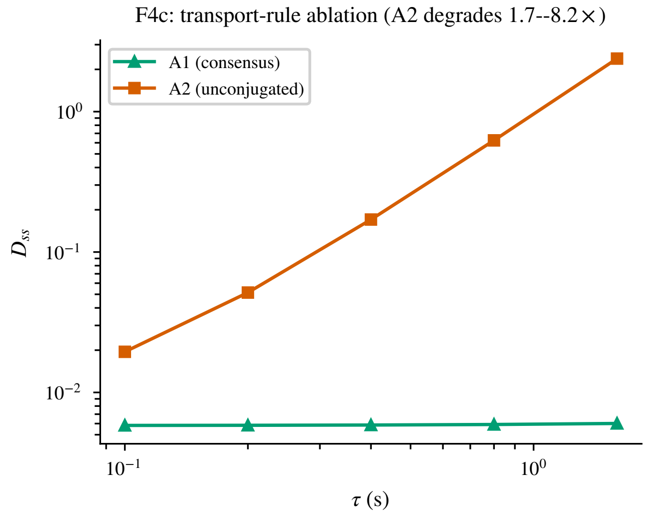

Fig. 20. All Drake arms on matched formation draws. Ablating the conjugated transport (A2) runs $1.7\times \to 8.2\times$ above the paper rule with growing τ; naive consensus (A1) holds disagreement flat and low—by groupthink (anchored ANEES 62, drift 8.4 m; baselines cell, 130 s horizon): agreement alone is never a virtue metric. The straight-tow control shows the motion excitation (19–21× at every τ).

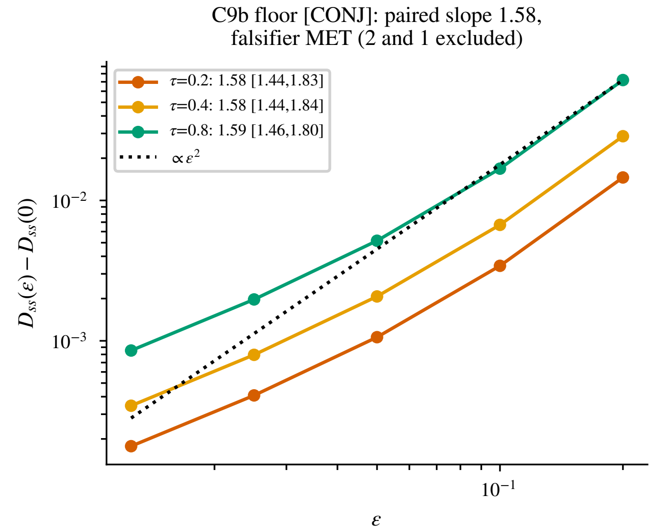

Fig. 22. Symmetry, the closed-loop floor [[CONJECTURAL]]: the seed-paired estimator gives slope 1.58 [1.44, 1.84] at every τ, excluding both 2 and 1 — a linear-plus-quadratic mixture the registered single-power model mis-specifies. Never mixed with the amplitude object.

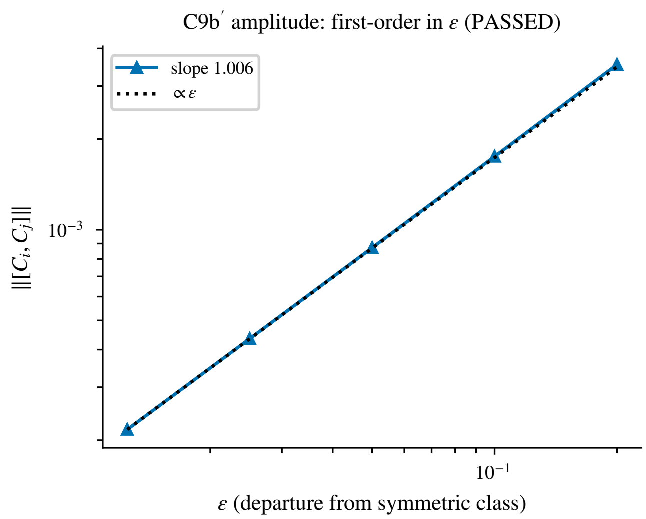

Fig. 21. Symmetry, the amplitude object: the departure from the symmetric class is first-order in ε (measured 1.006, base commutator exactly zero) — PASSED.

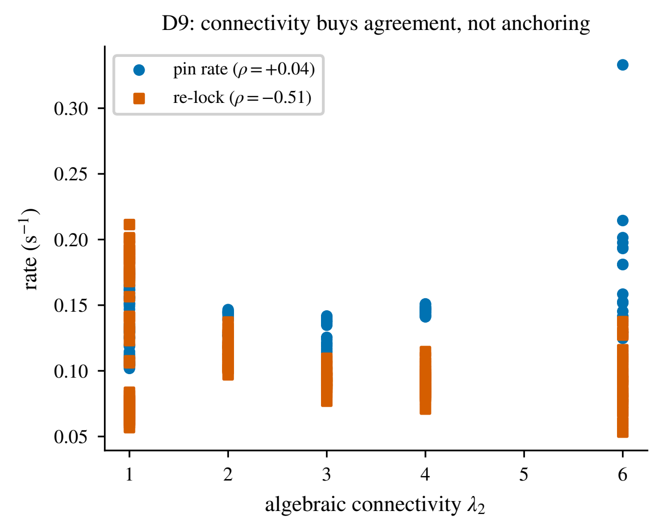

Fig. 23. Connectivity buys agreement, not anchoring. The pre-anchor floor improves with λ_2 (a); single-anchor pinning is anchor-rate-limited (b, Spearman $+0.04$); re-agreement around the pin *anti*-orders with connectivity (c, $-0.51\,[-0.65, -0.36]$). With the in-domain contraction claim verified (Fig. 24), this is the pre-registered ES-01 finding path, reported as such.

ment at high staleness); naive consensus buys flat agreement by groupthink and is exposed by every correctness metric; and the symmetric class's large protection (31×) lives entirely on the amplitude object, transferring only weakly (∼2–3×) to the closed-loop floor. Topology buys agreement, not anchoring: the floor improves with algebraic connectivity while single-anchor pinning is anchor-rate-limited, and re-agreement around a pin *anti*-orders with connectivity — consensus stiffness resists new absolute information, the same mechanism that makes naive consensus fail, appearing inside the record's own fusion at high degree.

c) What centralization is worth.: The strongest measured centralized reference — the record's own architecture at zero latency on the complete graph — improves accuracy by 6% at the calibration horizon and 19% at dock on long transits. One recorded re-architected centralized replay (B1-lite), granted zero latency and exact covariances, lost a factor of 2.2 to the decentralized record (0.41 vs. 0.183 m at the calibration horizon); a joint-EKF design run performed comparably (0.43 m, kept as a design record); and the granted-sensor baseline (B3) required six structural bring-up iterations (delay-consistent innovation, adjoint-transported residuals, joint geometric load

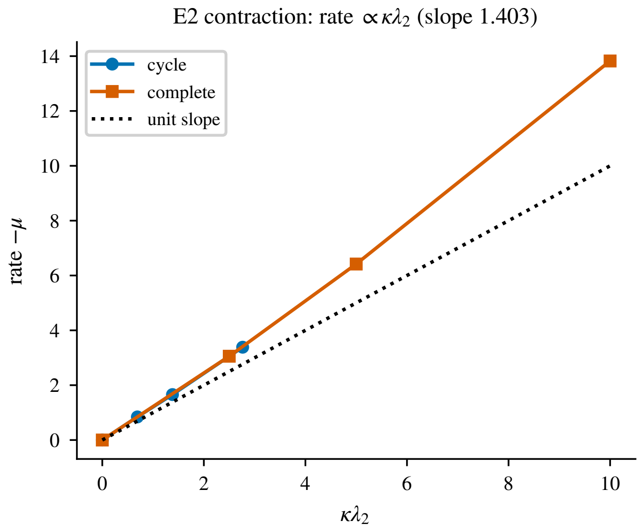

Fig. 24. Contraction in its own domain (frozen, unanchored): rate versus $\kappa\lambda_2$ across two topologies, slope 1.40 (falsifier < 0.5); μ calibrated at $\kappa = 0$, so the identity line is anchored, not predicted. The log-linearity remainder vanishes *faster* than the registered second order (2.48 [2.32, 2.65]; falsifier band excluded on the favorable side, reported as tripped with the corrected order).

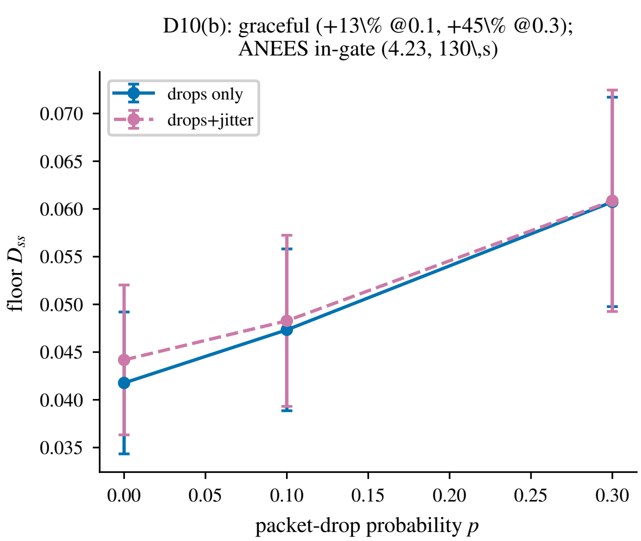

Fig. 25. D10(b) robustness (outside class $\mathcal{P}$): floor vs. packet-drop probability, drops-only and drops+jitter arms (8 seeds each; mean $\pm$ sd) — graceful, no red flags; anchored ANEES in-gate at p=0.3 at the 130 s horizon.

solve) without reaching competitiveness and is excluded as work in progress. We read this as evidence that the measurement architecture — per-agent Kalman-weighted constraint channels composed through conjugated transports — is where the accuracy lives, which is the design thesis under test.

d) Honest limits.: Figure 29 organizes the four adverse findings as a taxonomy — finding, measured cause, recorded remedy class, evidence file — so the limits are as structured as the results. The docking specification is unmet by every arm at plan-faithful beacon acquisition, including the zero-latency reference: the approach geometry, not the estimator, binds. A

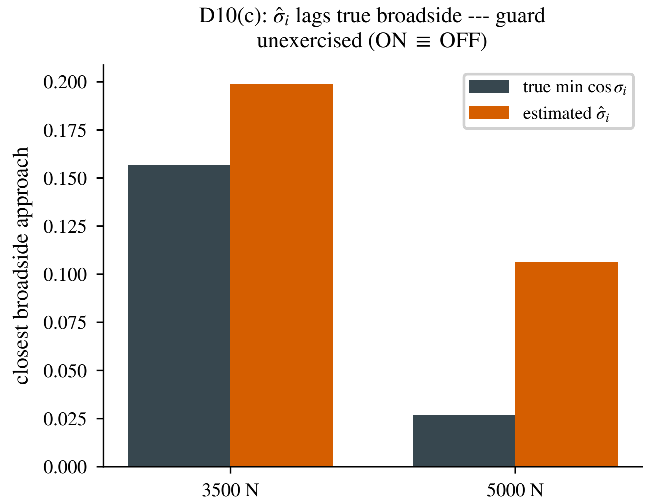

Fig. 26. D10(c): why the broadside guard never fired — $\hat{\sigma}_i$ lags true broadside across the reachable gust envelope, so ON $\equiv$ OFF. Reported as a null with its mechanism.

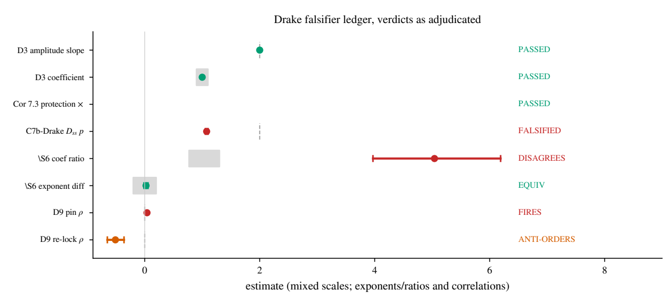

Fig. 27. The multibody falsifier ledger, verdicts as adjudicated: green passed, red falsified / fired, orange anti-ordered. Grey marks registered references and equivalence bands; evidence record named per row. [CONJECTURAL] rows report measured slopes as measured — no law asserted.

decelerating approach is necessary and was tested: it improves fleet-mean accuracy 2.4$\times$ (median 0.61 m) but leaves the spec unmet — the residual limit is anchored-agent gain starvation, in which sustained Kalman anchoring collapses the anchored covariance until the beacon gain starves, while CI's structural conservatism keeps the unanchored agents' gains healthy; the two covariance architectures fail in opposite directions on long horizons. A trim-contamination hypothesis was tested and falsified (frozen-trim runs bit-identical). The remedy class (covariance floors, adaptive beacon noise) is future work. The consistency gate closes only at the calibration horizon and does not transfer to 450 s transits; long-horizon bias growth exceeds the declared process noise, and every ANEES claim in this paper therefore carries its horizon. The broadside guard, gated on the estimated cable angle, lags true broadside during fast gusts and was never exercised in the reachable envelope — a lag-aware margin is the recorded design note. Slack-cable events are outside this plant's validity domain (bilateral distance constraints), scoped to future work with unilateral cables.

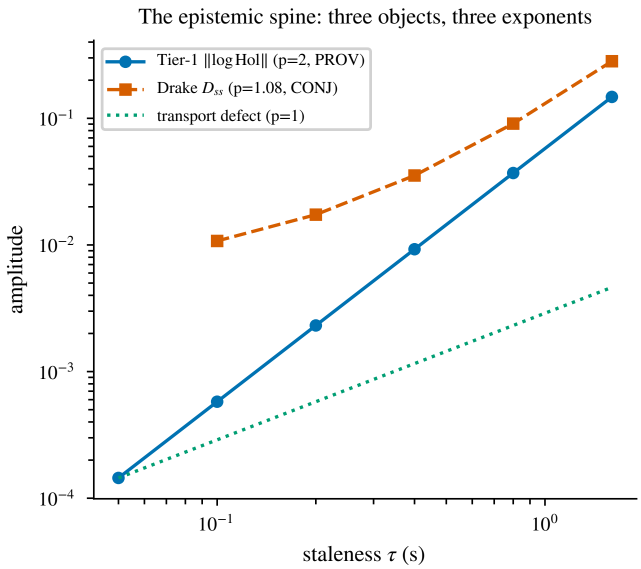

Fig. 28. The epistemic spine in one overlay: three distinct objects, three measured exponents — Tier-1 $\|\mathrm{Log\,Hol}\|$ ($p = 2.00$, the theorem's object, [PROVEN]); the Drake round-trip transport defect ($p = 1.00$, estimate-mismatch dominated); and the Drake closed-loop D_{ss} ($p = 1.08$ [1.05, 1.10], [CONJECTURAL] regime, D_0 measured by the straight-tow control 19–21× below). Never conflated.

TABLE IV
FAILURE TABLE: FREQUENCY, SEVERITY, DETECTION, MITIGATION.

failure	frequency	severity	detected by	mitigation
docking spec unmet	50/50 seeds, all arms (v2: 10/12)	task failure, recoverable	dock-window metric	decel approach (necessary); covariance remedy (revision)
anchored gain starvation	11/12 v2 seeds	degradation (anchored agent only)	per-agent error split	covariance floor / adaptive beacon R (revision)
ANEES non-transfer	all 450 s transits (159–229)	consistency claim limited	horizon-split ANEES	horizon-scoped claims (adopted)
guard unexercised	all paired gust runs (ON ≡ OFF)	latent (no harm observed)	envelope probes	lag-aware margin (design note)

IX. Supplementary Movies

Four supplementary movies ship with the paper; each is generated by a seeded, versioned script, and none is evidentiary — every quantitative claim traces to the committed campaign records, with the movies as faithful visualizations of recorded or re-runnable runs.

S1 (hero_dogleg_web.mp4). The full 450 s transit as a five-beat storyboard (9× time compression, badged on screen): gauge-orbit drift through GNSS denial, the turn bloom in $D(t)$, beacon pinning, and the honest docking scorecard as the closing card.

S2 (hero_blind_harbor.mp4). The gauge-pinning demonstration with ghost-caisson overlays: each agent's private load estimate drawn as a translucent caisson — coherently drifting, overlapping each other while offset from truth — and the $D(t)$ spike at acquisition, which is Cor. 5.2 live. (Playback ≈ 4.8× real time: stride-2 frames of the 10 Hz log at 24 fps.)

S3 (tow_N5.mp4). The physical scenario itself: the N=5 pentagon tow in the multibody plant — cables, drag, thrust allocation — so the reader sees the plant, not a diagram. (Playback 1.5× real time: 0.15 s log frames at 10 fps.)

S4 (error_anatomy.mp4). The failure-case movie, shipped deliberately: hero v2's fleet-mean *median* seed, per-agent errors alongside the scene. Pre-beacon, the fleet drifts *together* (the gauge, not disagreement); post-beacon, the unanchored agents converge through fusion while the anchored agent — covariance collapsed by sustained anchoring — starves its own beacon gain and stays high. The movie shows the paper's binding limit, not its best case. (Playback ≈ 25× real time: the 520 s transit in ≈21 s at 25 fps; the on-screen clock is authoritative.)

TABLE V
ARTIFACT REGISTRY (DRIVERS IN `tier2_drake/`; RECORDS IN `results/s1/`).

cell	driver
D2/D4+controls	`campaign/{production_d2_d4,d2_*}.py`
D3	`blind_harbor/d3_amplitude.py`
D6	`campaign/d6_tension.py`
D7	`campaign/d7_scorecard.py`, `campaign/d7_a2_arm.py`
D8	`campaign/d8_h_sweep.py`
D9	`campaign/d9_scaling.py`
D10(b,c)	`campaign/d10{b,c}_*.py`
B1/baselines	`campaign/{baselines,b1_true}.py`
hero v1/v2	`blind_harbor/hero_dogleg.py`, `campaign/hero_v2_ens`
movies	`blind_harbor/hero*.py`, `render.py` (S3), `campaign/erro`

X. Reproducibility and Artifact Registry

Every number in this paper traces to a seeded, versioned driver and a committed data file; `campaign_replay.py` re-executes per-run probes that must match the committed records exactly (Tier-1 to 10^{-9}; Drake to 10^{-6}), and `--run CELL` regenerates any full campaign. Table V maps each reported cell to its driver and record; the paper-figure generator (`analysis/figures/ral_artifacts.py`) reads *only* these committed records — it re-simulates nothing — so the figures cannot drift from the campaign. Movie generators are versioned beside the drivers; S4's underlying run is itself persisted (`error_anatomy_series.npz`).

XI. Research-Artifact Package

The paper ships as a structured evidence system rather than a set of plots: conceptual artifacts (the scenario and staleness ladder, Fig. 1; the linted architecture, Fig. 5); configuration artifacts (plant model and campaign matrix, Tables I–II); primary quantitative evidence (the scorecard and its distributions, Table III, Fig. 11; the recorded transit, Fig. 6; both-plant overlays, Figs. 18–20); causal attribution (ablations and symmetry, Figs. 21, 23; the v2 seed-level remedy test, Fig. 13); robustness (Fig. 25); statistical integrity (the adjudicated ledger, Fig. 27);

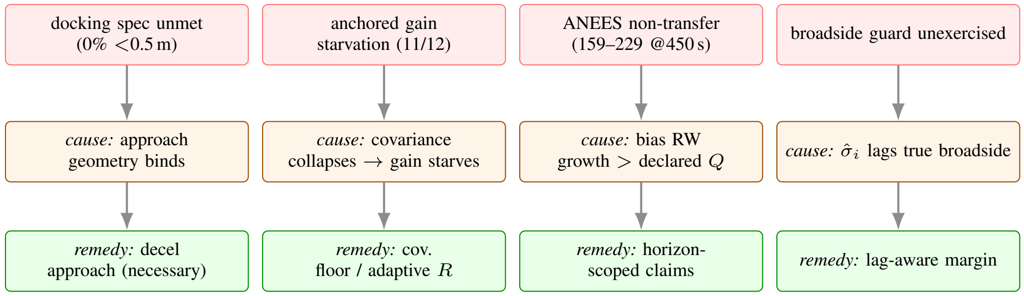

Fig. 29. Failure and limit taxonomy: each adverse finding with its measured cause and recorded remedy class. Table IV adds frequency, severity, and detection.

failure analysis (the taxonomy, Fig. 29; the per-agent anatomy, Fig. 10); four movies including a deliberate failure-case movie (S4); and the executable registry of Table V with its deterministic replay harness. Every adverse finding in the package carries its measured cause and its recorded remedy class; every ANEES number carries its horizon; and no disagreement metric is ever presented as a correctness ranking.

XII. Conclusion

On two independent plants sharing one estimator, the proven leading-order holonomy verifies to coefficient precision; the conjectured second-order closed-loop floor is falsified coherently, with the first-order estimate-mismatch channel dominating at realistic noise; and the sheaf-induced transport is the load-bearing element of the design — ablations, baselines, and the strongest measured centralized reference all lose to it or barely improve on it. The campaign's negative results are pre-registered, controlled, and localized; its remaining limits (docking approach, long-horizon calibration, guard lag, slack cables) are stated as measured boundaries with recorded remedies. All data, drivers, figures, and videos are versioned and exactly replayable.